CSA 1718 – Artificial Intelligenae

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Tower of Hanoi – Prolog

Aim

To write and implement a Prolog program to solve the **Tower of Hanoi problem using recursion**.

Objective

- To understand recursion in Prolog.
- To transfer disks from one peg to another.
- To follow the rules of the Tower of Hanoi problem.

Theory

The **Tower of Hanoi** consists of three pegs and a number of disks of different sizes. The objective is to move all disks from the source peg to the destination peg using an auxiliary peg.

Rules

1. Only one disk can be moved at a time.
2. Only the top disk can be moved.
3. A larger disk cannot be placed on a smaller disk.
4. All disks must be moved from the source peg to the destination peg.

Algorithm

1. If there is only one disk, move it directly from the source to the destination.
2. Move N-1 disks from the source to the auxiliary peg.
3. Move the remaining largest disk from the source to the destination.
4. Move the N-1 disks from the auxiliary peg to the destination.
5. Repeat the process recursively.

Program

```
hanoi(1, From, To, _) :-
    write('Move disk from '),
```

```
    write(From),
    write(' to '),
    write(To),
    nl.

hanoi(N, From, To, Aux) :-
    N > 1,
    N1 is N - 1,
    hanoi(N1, From, Aux, To),
    hanoi(1, From, To, Aux),
    hanoi(N1, Aux, To, From).
```

Query

For 3 disks, enter:

hanoi(3, left, right, center).

Output

Move disk from left to right

Move disk from left to center

Move disk from right to center

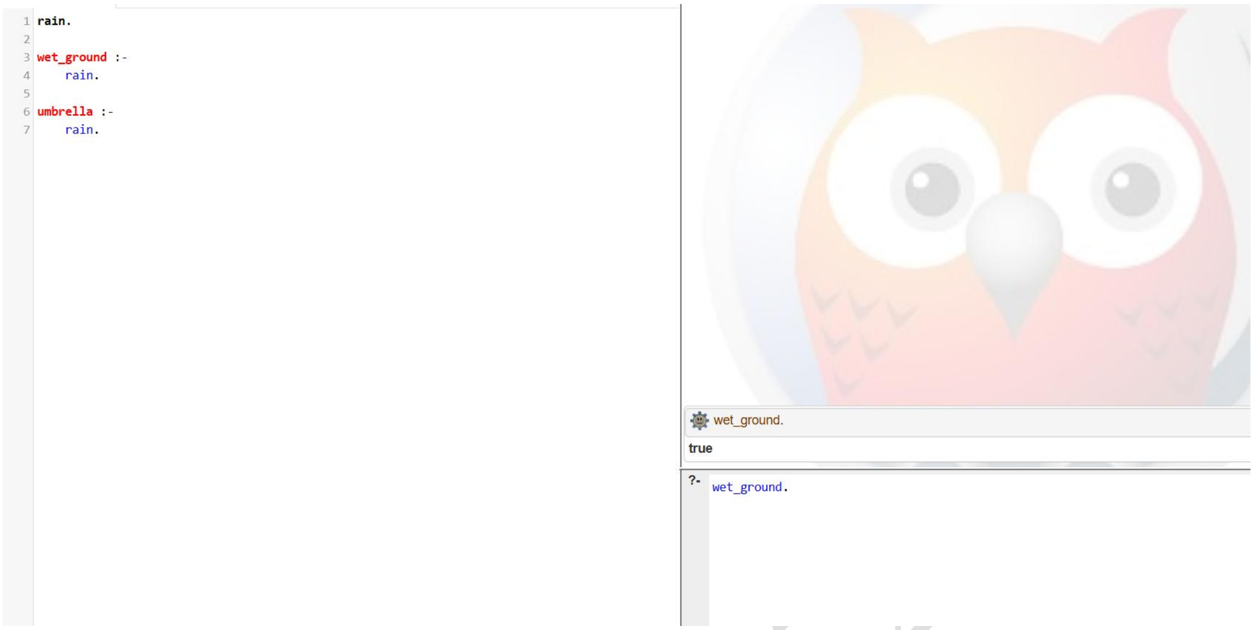
Move disk from left to right

Move disk from center to left

Move disk from center to right

Move disk from left to right

true.



Explanation

For 3 disks, the program performs **7 moves**.

The minimum number of moves is:

$\mathbf{2^N - 1}$

For 3 disks:

$\mathbf{2^3 - 1 = 7}$

Result

Thus, the **Tower of Hanoi problem was successfully implemented in Prolog using recursion**, and the required sequence of disk movements was obtained.